

Two Coarse-Grainings of One Arrow: General Relativity and Quantum Mechanics as Projections of a Discrete Relational Substrate

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Abstract

General relativity (GR) and quantum mechanics (QM) resist unification not because one is wrong but because each is a *lossy coarse-graining of the same thing*, and the two projections discard complementary information. This note makes that claim precise inside Event Density (ED), a framework in which spacetime and its fields are not fundamental but emerge from a discrete, relational substrate — a network of events, their participation, and their commitment — governed by thirteen primitives, one of which is an irreversible arrow of time. We show that both GR and QM are *time-symmetric*, so each has already coarse-grained the arrow out; what each retains is a different reversible residue of the substrate’s time, the local rate in GR and the global phase in QM. Because both are projections of one arrow, they are not unified by building a bridge between them but by recovering their common source. Concretely, ED identifies the object QM needs as its pointer-basis selector with the object GR needs as its preferred time: they are one primitive doing two jobs, and the standard “problem of time” therefore does not arise. We are explicit about tier: the QM side is a grounded reconstruction with a stated open residual, the weak-field GR side is derived, but the *unification itself is an identification at account tier* — distinctive, structural, and as yet carrying no falsifiable prediction of its own. We close by locating the familiar semiclassical regime as the domain where a conversion between the two time-notions exists — on an interpretive reading, gravitational time dilation is that conversion — and full quantum gravity as the regime where it must fail.

1. The reframe

For a century the unification of GR and QM has been posed as a bridge-building problem: quantize the metric, or geometrize the wavefunction, until the two formalisms meet in the middle. The program has been extraordinarily fertile and has not converged. ED suggests the framing is the obstacle.

The ED claim is that GR and QM are **two coarse-grainings of a single substrate**, and that they cannot be glued because each threw away part of the substrate that the other kept. On this view unification is not a bridge across two theories but a *decompression down* to the layer both are shadows of. The layer is a discrete relational substrate whose one indispensable primitive is the arrow of time. GR and QM are what you get when you read that substrate at the continuum scale in two different, mutually blind ways.

This note is self-contained. Section 2 introduces the substrate and its primitives. Sections 3 and 4 recall, at their stated tiers, what the arrow does in QM and in GR. Section 5 makes the reframe, Section 6 discusses the conversion between the two time-notions and where it breaks, and Section 7 states plainly what is and is not being claimed.

2. The Event Density substrate

ED does not begin with spacetime, a metric, or an action. It begins with a substrate that is:

- **Discrete.** The fundamental furniture is events, not points of a continuum. There is a minimum length, the grain, identified with the Planck length.
- **Relational.** An event is nothing except through its participations. Two is more fundamental than one: an isolated element carries no structure. Space, geometry, and fields are patterns of which events participate with which, not a stage the events sit on.
- **Arrowed.** The substrate advances by *commitment*, an irreversible, non-invertible (many-to-one) step. Time's direction is written into the update rule itself, not recovered afterward from a boundary condition.

The substrate is governed by thirteen primitives, the theory's entire axiomatic input (Primitive tier throughout):

- **P01** an existence layer; **P02** participation as the primitive relation; **P03** channel-and-locus indexing with spatial homogeneity; **P04** a non-negative, additive scalar bandwidth; **P05** polarity-transport between adjacent loci; **P06** spatial dimension $D = 3+1$; **P07** channel structure; **P08** the substrate scale $\ell_{ED} = \ell_P$; **P09** $U(1)$ -valued polarity; **P10** a rule-type primitive that licenses cross-chain memory; **P11** commitment, irreversible; **P12** a stability functional $\Sigma = \text{Coh} - \text{Str} - \text{Grad}$; **P13** time-translation homogeneity.

Two of these carry the present argument. **P11** is the arrow in its raw form: the primitive content is non-invertibility, the update from now to next is many-to-one, so the past cannot be read back off the present uniquely. **P13** fixes time-translation homogeneity — no moment is privileged — which is what makes P11's commitment-direction the *only* temporal-asymmetry primitive; the discreteness of events, and hence a countable sequence of substrate ticks, belongs to the substrate itself (P01), not to P13. The directional arrow of everyday experience is not P11 alone; it is the derived result T18, in which P11, P13, and the single-chain memory kernel V1 jointly fix time's direction. V1 is a derived kernel, not a primitive; its strictly-retarded support is part of what T18 establishes (Paper_089). Cross-chain correlations are carried by a second kernel, V5, licensed by P10. For this paper the essential fact is simply that **irreversibility is primitive, not emergent**.

Continuum physics is reached from the substrate by coarse-graining. The Discrete-to-Continuum Grounding Theorem (DCGT) is ED's bridge: continuum fields and their equations of motion are what survive when substrate detail is averaged away in a controlled limit. A structural consequence, used repeatedly below, is that **continuum partial differential equations are coarse-graining artifacts**. They can carry *more* symmetry than the substrate they came from. In particular, a reversible continuum law is not evidence that the substrate is reversible; time-reversal symmetry is one of the things a coarse-graining can manufacture by discarding the arrow.

3. The arrow in quantum mechanics: it selects the pointer basis

ED reconstructs quantum kinematics from the substrate rather than postulating it. The load-bearing move (the Gleason/quantum-logic keystone) reduces orthogonality to operational distinguishability, selects the complex field $\mathbb{C}$ from the substrate's phase and composition structure, and grounds the Born rule as a participation-frequency limit. Unitary evolution appears as the coarse-grained dynamics of coherent participation, yielding the Schrödinger equation via Stone's theorem; the uncertainty relation and the standard operator apparatus follow, all resting on the same reconstructed inner product whose Solèr-lattice-rigor step is the acknowledged open residual (below).

The piece relevant here is measurement. In ED a measurement is a *commitment*: an irreversible, many-to-one substrate step. Because commitment happens in the channel basis and not in phase, **the channel basis is the unique pointer basis, selected by the arrow**. This is einselection, and in ED it is *primitive* rather than emergent from environmental decoherence: the arrow fundamentally picks the classical basis, and the familiar basis-free, multi-context QM is what emerges at the apparatus level. Repeatability of measurement follows from the same irreversibility that locks the committed record.

Tier: the QM reconstruction is **grounded / reconstructed**, with one honest residual — the full Solèr-lattice rigor step remains open. What is solid independent of that residual is the reframe: orthogonality is operational, $\mathbb{C}$ is selected, and **the arrow selects the pointer basis**. Hold that identification; Section 5 needs it.

4. The arrow in gravity: it is the preferred time

On the gravitational side ED recovers, at the weak-field level, the standard results by treating the metric as an emergent, coarse-grained reading of the substrate's local commitment structure. A temporal lapse $N^2 \propto b$ (with b the local bandwidth) yields the weak-field Einstein metric on the Schwarzschild branch, the factor-of-two light bending, and the field-equation form; these are **derived** at that level (GR-I, GR-II). The emergent theory is *khronometric*: two tensor modes plus one khronon scalar, with the gauge group the foliation-preserving diffeomorphisms. The khronon is not a new field standing beside the arrow. **The khronon is the arrow made dynamical** — the substrate's global direction of commitment, promoted to a field. In the deep-field (low-acceleration) regime the same khronon reproduces the MOND phenomenology with a parameter-free acceleration scale $a_0 = cH_0/2\pi$ (KM-I; galactic passes are grounded, values inherited).

Crucially, ED's gravity carries a genuine preferred time, and it survives observation because its signature is buried. The preferred-frame post-Newtonian parameters are $\alpha_2 = 0$ exactly (structural) and α_1 suppressed by at least seventy orders of magnitude, because commitment is forced sparse (GR-IV). The preferred frame exists; its effects are simply unobservable at present precision.

Tier: weak-field GR is **derived**; the galactic sector is **grounded** with inherited values; the dark-energy magnitude is inherited via H_0 . Hold the identification: **gravity's preferred time is the arrow**.

5. Why they will not glue, and how they unify

Now place the two identifications side by side. In QM the arrow is the pointer-basis selector. In GR the arrow is the preferred time. Section 6 of the flagship Report makes the unifying observation: these are not two facts. **They are one primitive — the arrow — doing two jobs.** That is the unification, and it is an *identification*, not a new mechanism introduced to bolt the sectors together.

The reframe explains the century of difficulty directly. Consider what GR and QM each did to time:

- **Both are time-symmetric.** Einstein's equations run backward as readily as forward; unitary Schrödinger evolution is reversible. So each theory has *already coarse-grained the arrow out*. Neither continuum law retains the irreversibility that the substrate has as a primitive. This is exactly the DCGT lesson of Section 2: the reversible continuum symmetry is a coarse-graining artifact, not a property of the base.
- **Each keeps a different reversible residue of the substrate's time.** GR keeps the *local rate*: the lapse, how fast proper time ticks here versus there. Its time is duration, geometric and position-dependent. QM keeps the *global phase order*: the single background parameter t against which every amplitude's phase advances as $e^{(-iHt/\hbar)}$. Its time is coherent ordering, uniform and background.

Two lossy shadows of one object, and they discarded complementary parts. GR's time has no phase; QM's time has no local geometry. You cannot glue two projections that each deleted what the other saved, and you certainly cannot rebuild the whole from two pieces that both deleted the load-bearing part, the arrow. The bridge-building program was trying to reconstruct a primitive by welding together two things that had each removed it.

ED's move is the opposite of a bridge. Put the arrow back at the base and both time-notions fall out as its two coarse-grainings: project the arrow one way and you get the lapse GR reads; project it another way and you get the phase parameter QM reads. **The link was never a bridge between them; it is their common source.** In this precise sense the two theories are threads of one rope, or projections of one substrate reality.

The immediate payoff is that the **problem of time dissolves rather than getting solved**. The problem of time is the standard obstruction that GR admits no preferred time while canonical quantization produces a frozen, timeless constraint. ED does not split the difference: it sides with quantum mechanics in holding that there *is* a fundamental preferred time, the arrow, and then observes that this same object is what gravity's khronon foliation picks out. A recognized tension is made not to arise. That is real work, and it is worth being clear it is *dissolution*, not proof.

6. The conversion between the two times, and where it breaks

The reframe also locates, and explains the limits of, the intuition that one ought to be able to *convert* between GR-time and QM-time. There is such a conversion, and it is already familiar.

Write the substrate's uniform tick count as n , the P13-homogeneous ticking of the substrate. Commitments in the sense of P11 are the *sparse* events occurring within these ticks; they do not fire every tick, and that sparseness is exactly what suppresses α_1 by ≥ 70 orders in GR-IV. QM's

parameter t is, to leading order, this uniform tick count: phase advances as $e^{\hat{(-iHn)}}$. GR's proper time is the *same* ticks weighted by the local commitment density, $\tau = \int (\text{local rate}) \, dn$, with the local rate the number of commitments per tick. Their ratio,

$d\tau/dt = \text{local rate of becoming} = \text{the lapse},$

is nothing other than gravitational time dilation. **The lapse is the conversion factor between the QM and GR readings of time.**

This conversion is available precisely when the lapse can be held as a fixed classical background. That regime has a name: quantum field theory in curved spacetime, quantum fields (QM-time) evolving on a frozen gravitational lapse (GR-time). It is exactly where the semiclassical results, Hawking and Unruh radiation among them, are computed, and it works. The conversion *fails* precisely when the lapse itself must become a dynamical, fluctuating substrate quantity, which is full quantum gravity. In substrate terms: the two projections are jointly faithful wherever one side may be treated as a fixed background, and there is no clean conversion when the background is itself a live substrate observable. The intuitive “can't mash them” is, on this reading, the statement that the conversion factor stopped being a fixed number and became a fluctuating field. This section is interpretive: it reframes standard semiclassical gravity through the substrate lens; it is not a new derived result.

7. Scope and status: what this is, and what it is not

This paper composes existing ED results into a cold-reader argument. It is important, given the subject, to be exact about tier.

What is claimed, and at what tier. - The ED substrate and its thirteen primitives are **postulated axioms** (Primitive tier). The irreversibility of commitment (P11) is primitive; the directional arrow is derived from it (T18). - The QM side — the arrow selects the pointer basis, einselection primitive, Born rule and companions — is a **grounded reconstruction** with the Solèr-lattice-rigor step an acknowledged open residual. - The weak-field GR side — khronometric metric, khronon = dynamical arrow, preferred-frame invisibility — is **derived** (weak field) and **grounded** (galactic), with dark-sector magnitudes inherited. - The unification itself — that the pointer-basis selector and the preferred time are one primitive — is an **identification at account tier**. It is distinctive and structural. It is *not* forced by a deeper theorem, and it delivers *no falsifiable prediction of its own*. - The coarse-graining/projection narrative (Sections 5–6), including the lapse-as-conversion reading, is an **interpretive lens** consistent with the derived and grounded results, not an independent derivation.

What is not claimed. ED does not here provide a quantized graviton, a Hilbert-space unitarity proof, or a UV completion in the usual sense; the substrate's discreteness reframes UV questions rather than answering them in the continuum's language. It does not claim to have *proved* that GR and QM are unified. It claims to have *dissolved* the problem of time by identifying one primitive behind two demands, and to have located why the semiclassical conversion between the two time-notions exists and where it must fail.

Why it is nonetheless substantive. A framework in which the arrow is primitive predicts the *structure* of the difficulty: because both continuum theories are time-symmetric, both must have discarded the arrow, so neither can host the other, and unification cannot live at the continuum level at all. That the mainstream obstruction (the problem of time) is exactly a

conflict about time, and that it evaporates the moment the arrow is restored to the base, is the kind of retrodiction a serious reframe should produce. It is offered as such, honestly tiered, and with its one genuine gap (Solèr rigor) and its central caveat (account, not theorem; no standalone prediction) stated in the open.

8. Conclusion

GR and QM are not two theories waiting to be joined. They are two coarse-grainings of one arrowed relational substrate, each keeping a complementary reversible residue of the substrate's time and each blind to the other's. They unify not by a bridge but by their common source, and the object that does the unifying is the same primitive that selects the quantum pointer basis and fixes gravity's preferred time. Restore the arrow to the base and the problem of time does not arise. The holy grail, on this reading, was never a bridge to build; it was a source to recover.